

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2023
MATH-1105: Differential and Integral Calculus

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Explain increasing and decreasing function with graph. 3
 b) Test the Continuity and Differentiability of the following function at $x = \frac{\pi}{2}$ 6

$$f(x) = \begin{cases} 1 & \text{when } x < 0 \\ 1 + \sin x & \text{when } 0 \leq x \leq \frac{\pi}{2} \\ 2 + (x - \frac{\pi}{2})^2 & \text{when } x \geq \frac{\pi}{2} \end{cases}$$

 c) Sketch the graph of $f(x) = 2x^5 - 5x^2 + 1$. 5

2. a) Draw the rough graph of the following: 5
 i) $y = |x+2|$; ii) $(y+2)^2 = x$; iii) $y = \sin x$; iv) $y = e^x + 1$.
 Use vertical line test to determine whether the above equations are function or not. 5
 b) What do you mean by ' $f(x)$ is continuous at $x = a$ '?
 Let $f(x) = \begin{cases} x-1, & x \leq 3 \\ 3x-7, & x > 3 \end{cases}$, Find i) $\lim_{x \rightarrow 3^-} f(x)$; ii) $\lim_{x \rightarrow 3^+} f(x)$; iii) Does $\lim_{x \rightarrow 3} f(x)$ exist?
 Determine whether $f(x)$ is continuous or not at $x = 3$? 4
 c) Determine whether $f(x) = \begin{cases} x \cos \frac{1}{x}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$ is differentiable or not at $x = 0$? 6

3. a) Find the extremum values of 8

$$f(x) = \frac{x^3}{3} + ax^2 - 3a^2x, a < 0.$$

 Also find the critical points, if any, of the above function.
 b) Find the intervals on which $f(x)$ are (i) increasing; (ii) decreasing; (iii) concave up; 8
 (iv) concave down, where $f(x) = 5 + 12x - x^3$. Also find the inflection points, if any, of the above function.

4. a) Describe Linear Approximation with examples. Find the linear approximation of 6
 $f(x) = \sqrt{x}$ at $x = 9$ with graph. 3
 b) Find $(f \circ g \circ h)(x)$ if $f(x) = \sqrt{x}$, $g(x) = \frac{1}{x}$ and $h(x) = x^3$. 5
 c) The value of a car is Tk. 21500. It losses 15% of it's value every year. 7
 i) Write a function that represents the value (in Tk.) of the car after t years.
 ii) Find the approximate monthly percent decrease in value.

5. a) Prove that $\Gamma \frac{1}{2} = \sqrt{\pi}$ 3
 b) Evaluate (Using L'Hospital rule) 4
 i) $\lim_{x \rightarrow 1} (\frac{x}{x-1} - \frac{1}{\ln x})$
 ii) $\lim_{x \rightarrow 0} \frac{x - \tan x}{x^3}$
 c) Find the Volume of the solid generated by revolving the area enclosed between the 7
 curve $y^2 = 2x$ and the straight line $y = 3x$ about X axis.

6. a) Integrate any TWO of the following function with respect to x :

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i) $f(x) = \frac{e^{\cot^{-1} x}}{(1+x^2)^2}$; ii) $g(x) = e^{ax} \sin bx$; iii) $h(x) = \frac{1+x^{\frac{1}{2}}}{1+x^2}$.

- b) What is the physical meaning of

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$\int_a^b f(x) dx$? Evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$.

- c) Show that

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$$\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(x) \text{ is even} \\ 0, & \text{if } f(x) \text{ is odd} \end{cases}$$

7. a) Find the Surface area of the solid bounded by the revolution of the cycloid $x = a(\theta - \sin \theta)$ and $y = a(1 - \cos \theta)$.

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- b) Find the volume of the solid formed by the revolution of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about X axis.

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